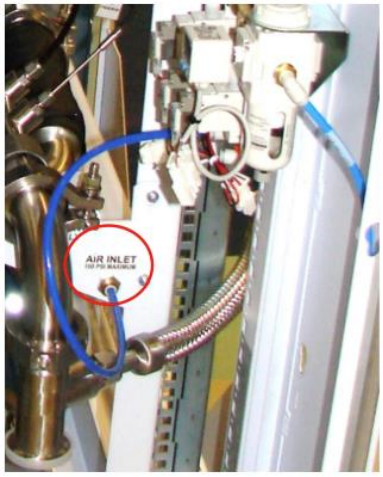
Build Overview:

1 **CDU Support Bracket Install**
2 **Power Conduit Install**
3 **PSU HFE Connections to CDU**
4 **Air Regulator Install**



CHANGE HISTORY

Date	BPM Rev	Change Details	Updated by
	1	New Issue	JOSEPH AMAD

Prepared By:
JOSEPH R. AMAD
Manufacturing Engineer

Checked By:
IVAN NOWELL PANIQUE
Team Lead

Approved By:
SHERWIN ARIOSTE
Operations Manager

MAN: CERTIFIED OPERATOR, TECHNICIAN

MATERIAL:

ITEM NO.	PART NUMBER	DESCRIPTION	QTY.
1	472-426-20	M5 Screws	19
2	631-313-12	Power Pipeline	1
3	470-302-00	Grommet	3
4	631-314-00	Upper Cover	1
5	634-442-00	Signal Line	1
6	634-437-00	Ground Wire	1
7	813-929-01	Tie	3
8	361-882-00	Clip	4
9	640-247-00	Ball Head Gasket	4
10	470-778-04	Spring Cushion	4
11	470-022-85	Hex Nuts	4
12	633-220-00	T Tube	2
13	360-557-08	Pipe Fittings	2
14	638-030-00	Molykote Grease	1
15	361-883-00	90° elbow	2
16	361-886-00	Washer	3
17	634-569-00	Bracket	1
18	614-766-90	M8 Frame Locking Block	2
19	470-933-01	M8 Screw	2
20	470-502-07	M8 Spring Pad	2
21	470-501-06	M8 Flat Washer	2
22	653-632-00	Regulator	1
23	637-717-00	Dixon Connector	1
24	637-746-01	560 Trachea	1
25	634-441-00	CDU Chain Line	1
26	637-746-00	500mm Trachea	1

MACHINE

NO.	PART # / ID	DESCRIPTION	IMAGE
1	N/A	PH2 Phillips Torque Screwdriver 3.3 N.m	N/A
2	N/A	17mm Socket Wrench 5.6 N.m	N/A
3	N/A	13mm Socket Wrench 3.3 N.m	N/A
4	N/A	15/6 open torque wrench 6.8 N.m	N/A
5	N/A	T20 Torx Torque Screwdriver 1.6 N.m	N/A
6	N/A	8mm Socket Wrench 1.6 N.m	N/A
7	N/A	1 Inch Open End Wrench	N/A
8	N/A	5mm Hex Bit 13.5 N.m	N/A

MAN: :CERTIFIED OPERATOR, TECHNICIAN

MATERIAL: N/A

MACHINE: N/A

METHOD:

3. Operation steps

Install CDU 632-199-01

1.1 As shown in the figure, use 4 M5 screws (472-426-20) to fix the CDU on the guide rail on the operating surface of the cabinet. The tool is a PH2 Phillips torque screwdriver. The torque value is 3.3 N·m.

1.2 Fix the CDU on the beam with two M5 screws (472-426-12) on the maintenance surface of the cabinet as shown in the figure. The tool is a PH2 Phillips torque screwdriver. The torque value is 3.3 N·m.

1.3 Tighten the 4 previously unlocked M5 screws on the CDU "Z" bracket as shown in the figure, the tool is a PH2 Phillips torque screwdriver, The torque value is 3.3 N·m.

MAN: :CERTIFIED OPERATOR, TECHNICIAN

MATERIAL: N/A

MACHINE: N/A

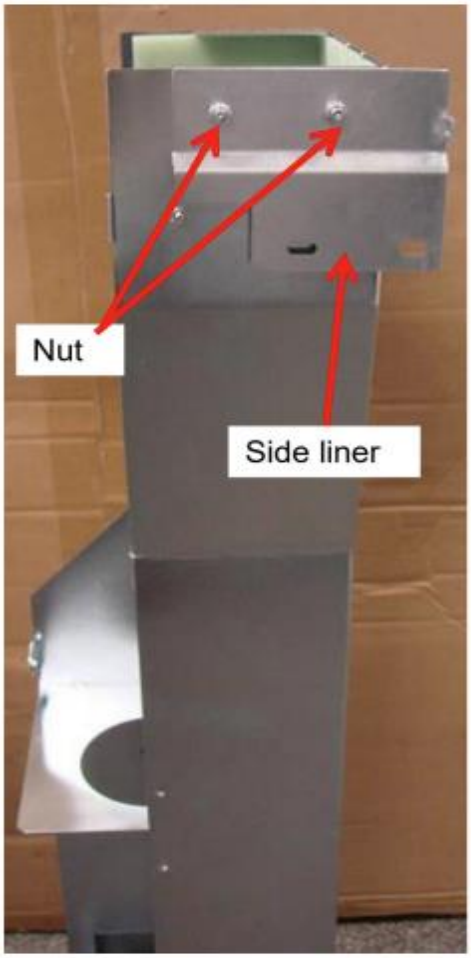
METHOD:

Installation of wire ducts

2.1 Remove the indicating screw from ADB, save the screw and use it again.

(Tool is T20 plum blossom head)

Take a power pipe 631-313-12, remove the two nuts above, and remove the side liner, save the nut and side liner for later use. (Tool 8mm socket wrench)



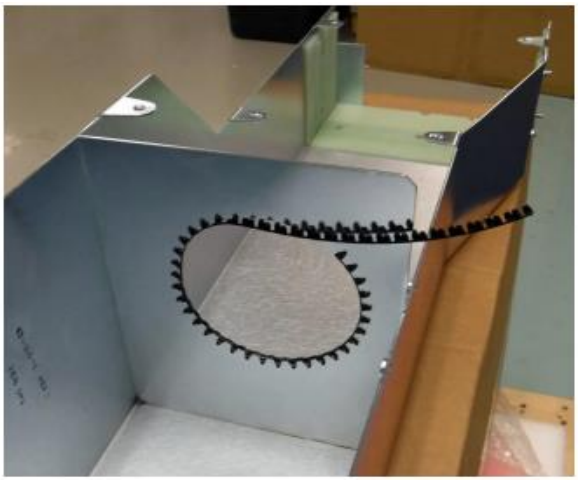
MAN: :CERTIFIED OPERATOR, TECHNICIAN

MATERIAL: N/A

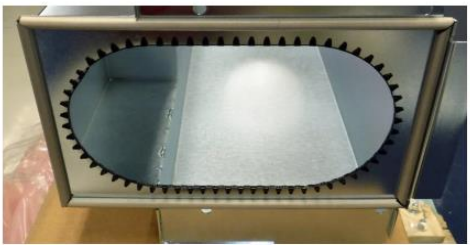
MACHINE: N/A

METHOD:

2.2 Place the wire conduit 631-313-12 on the desktop with the opening facing upward:
2.3 Install a grommet 470-302-00 in the middle round hole. The grommet fits the round hole and cut off the excess.



2.4 As shown in the figure, install the grommet 470-302-00 in the oval hole at the bottom of the wire duct, and the grommet can be adapted to the hole.
Need to use 2 grommet 470-302-00, install one first, then connect the second one to the first one, complete the cutting after the set is complete.



MAN: :CERTIFIED OPERATOR, TECHNICIAN

MATERIAL: **N/A**

MACHINE: **N/A**

METHOD:

2.5 Install the wire duct (631-313-12) with grommet between the cabinet maintenance surface bottom plate and ADB as shown in the figure. If necessary, lift slightly. ADB then puts the wire holder in the right position.

2.6 Use 4 M5 screws (472-426-12) to fix the bottom of the cabinet as shown in the figure. The tool is a PH2 Phillips torque screwdriver with a torque value of 3.3 N·m.

MAN: :CERTIFIED OPERATOR, TECHNICIAN

MATERIAL: N/A

MACHINE: N/A

METHOD:

2.7 Install the upper cover plate 631-314-00 on the wire duct as shown in the figure.
Use 9 M5 screws (472-426-12), do not tighten them, just screw them on.

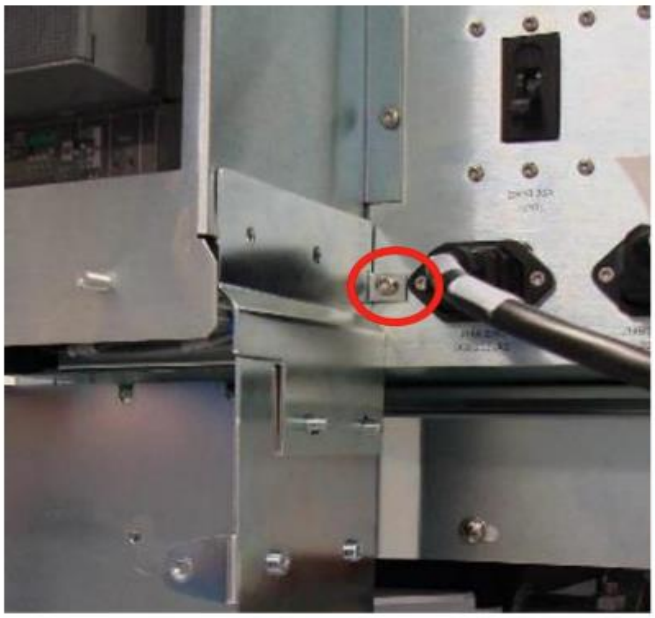
MAN: :CERTIFIED OPERATOR, TECHNICIAN

MATERIAL: N/A

MACHINE: N/A

METHOD:

2.8 Reinstall the side panel as shown in the figure, and tighten it with the screws reserved before ADB,Tool is T20 Torx screwdriver, The torque value is 1.6 N·m.
Then tighten the two nuts that were reserved before,Tool is 8mm socket wrench, The torque value is 1.6 N·m.



MAN: :CERTIFIED OPERATOR, TECHNICIAN

MATERIAL: **N/A**

MACHINE: **N/A**

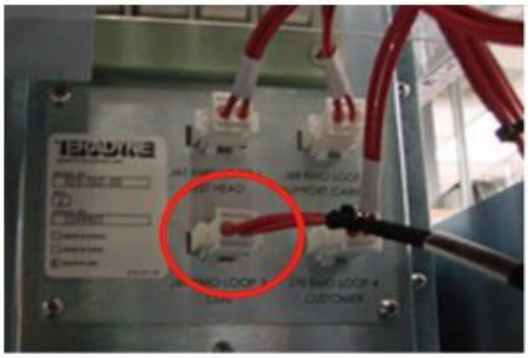
METHOD:

3.1 Take a wire (634-442-00) as shown in the figure, connect one end to the ADB-P69 interface, and the other end to the CDU-J2 interface.

3.2 Take a wire (634-437-00) as shown in the figure, connect one end to the ground panel Mecca (installed in the previous step), and connect the other end to the CDU ground port.

Use one M6 nut (603-096-03), the tool is a 10mm socket wrench, and the torque value is 5.6 N·m.

Note: If the CDU control line is uninstalled during transportation, reinstall it to J6 now.



MAN: :CERTIFIED OPERATOR, TECHNICIAN

MATERIAL: N/A

MACHINE: N/A

METHOD:

3.3 Insert the CDU power cord into the J10 interface as shown in the figure. Press the power cord in and turn it clockwise to screw in.

3.4 As shown in the figure, tie the cable to the guide rail with 2 cable ties (813-929-01).

MAN: :CERTIFIED OPERATOR, TECHNICIAN

MATERIAL: N/A

MACHINE: N/A

METHOD:

4.1 Reconfigure 4 clips (361-882-00) as shown in the figure:

a. Remove and discard the nut.

b. Load the following materials into the screw:

1.640-247-00 ball head gasket.

2.470-778-04 spring cushion.

3.470-022-85 hex nut.

MAN: :CERTIFIED OPERATOR, TECHNICIAN

MATERIAL: N/A

MACHINE: N/A

METHOD:

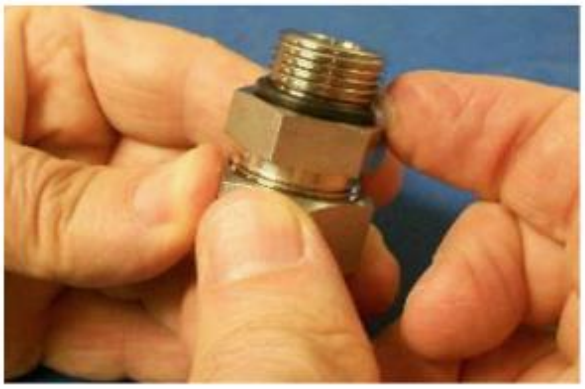
4.2 Take two T-tubes "633-220-00", two pipe fittings (360-557-08) and Molykote grease (638-030-00)

a. Apply Molykote grease to the O-ring of the pipe fitting.

b. Put the pipe into the T-shaped pipe and fix it with a 15/16 open torque

The torque value is 6.8 N·m.

c. Repeat the above operation on the second T-tube.



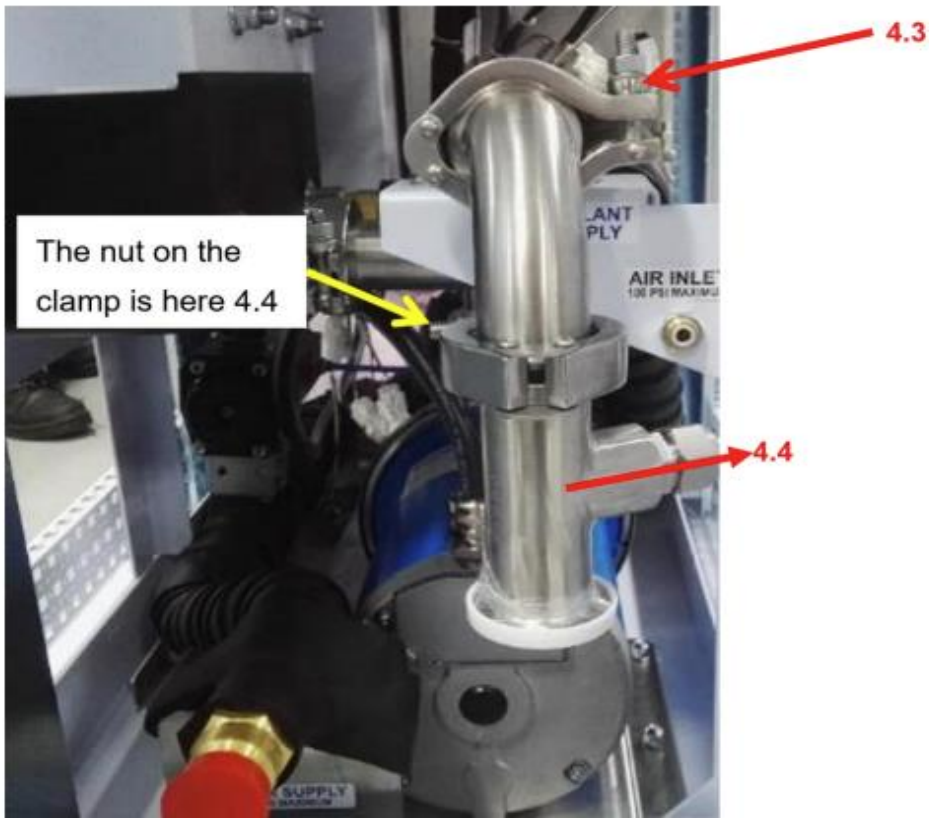
MAN: :CERTIFIED OPERATOR, TECHNICIAN

MATERIAL: N/A

MACHINE: N/A

METHOD:

4.3 Install the 90° elbow (361-883-00) on the CDU cooling supply pipe as shown in the figure.
Use 1 clip (361-882-00) and washer (361-886-00) fixed.
Note: The outlet direction of the outlet 90° elbow is downward as shown in the figure.
The direction of the nut on the clamp is toward the upper right position.
4.4 Install the T-shaped pipe (633-220-00) on the previous 90° elbow as shown in the figure.
Use 1 clip (361-882-00) and washer (361-886-00) fixed.
Note: The exit direction of the T-tube is to the right as shown in the figure. The direction of the clamp nut is toward the left position on the back.
Note: Do not use torque to fix the clip. The T-shaped tube needs to be loosened when the tube is installed in the T-shaped fitting.



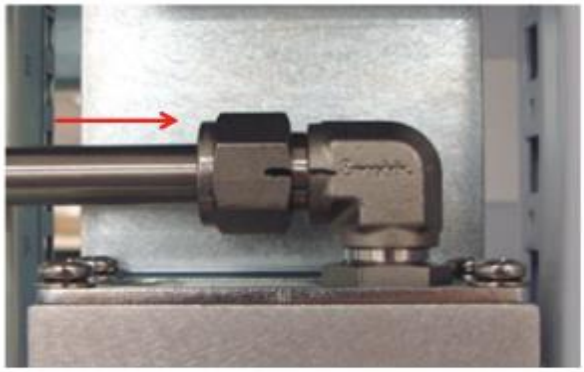
MAN: :CERTIFIED OPERATOR, TECHNICIAN

MATERIAL: **N/A**

MACHINE: **N/A**

METHOD:

Internal components of the joint sleeve.
a. Pay attention to the orientation of the nut and 2 metal ferrules as shown in the figure.
The key is to remove the metal ferrule from the nut and install it in the order and direction shown in the figure.
The installation process of the joint casing.
a. Insert the metal pipe firmly into the casing joint to ensure full insertion and manually tighten the casing joint.
b. Mark the metal joints and nuts as shown in the figure.
c. Use a 1-inch open-end wrench to tighten the nut to 1-1/4 turns.



MAN: :CERTIFIED OPERATOR, TECHNICIAN

MATERIAL: **N/A**

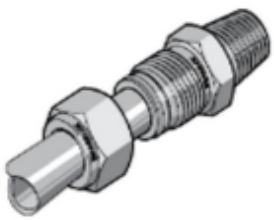
MACHINE: **N/A**

METHOD:

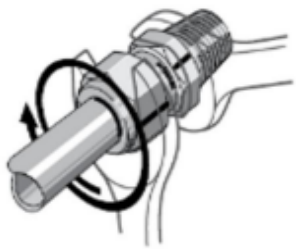
Reassemble the joint sleeve. (Use this page for rework or secondary assembly of joint casing)
You may disassemble or reassemble the joint casing several times. Please use this page to install.
Note: Before removing the joint casing part, it must be ensured that the system has been decompressed.

Before disassembly, make a mark and draw a line on the nut and sleeve joint at the back of the pipe.

The purpose of making these marks is to ensure that the nuts can be returned to their original positions when re-assembled and screwed back.



Insert the pipe with the pre-installed metal ferrule into the sleeve joint until the front metal ferrule is against the joint.



While using a wrench to clamp the casing joint, use another wrench to rotate the nut to the previously tightened position and keep it flush with the mark made before. At the same time, you will feel the resistance increase significantly, tighten the nut gently.

MAN: :CERTIFIED OPERATOR, TECHNICIAN

MATERIAL: N/A

MACHINE: N/A

METHOD:

5.1 Insert the hose into the T-tube as shown.

a. Rotate the 90° elbow and T-tube direction as required so that the hose can be inserted into the pipe fitting.

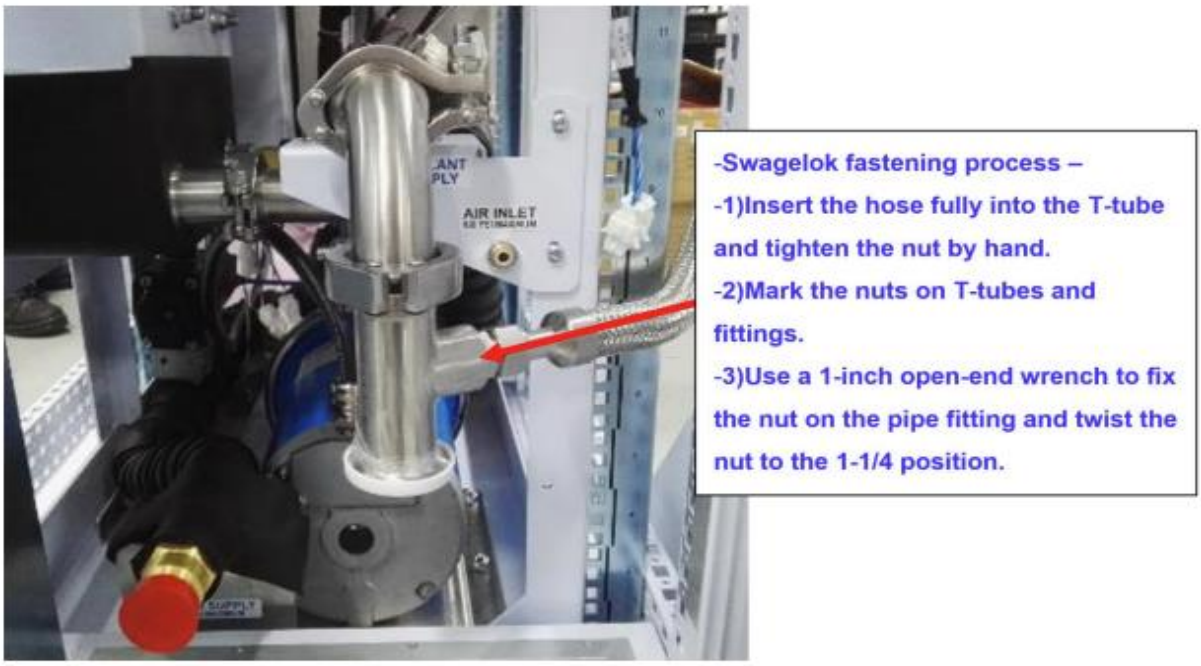
b. Confirm that the hose is fully inserted into the fitting.

5.2 Manually tighten the nuts on the pipe fittings.

5.3 Use a 13mm socket wrench to fix the nut on the clamp with a torque value of 3.3 N·m.

5.4 Mark the nuts on T-tubes and fittings.

5.5 Use a 1-inch open-end wrench to fix the nut on the pipe fitting, and twist the nut to the 1-1/4 position.



MAN: :CERTIFIED OPERATOR, TECHNICIAN

MATERIAL: N/A

MACHINE: N/A

METHOD:

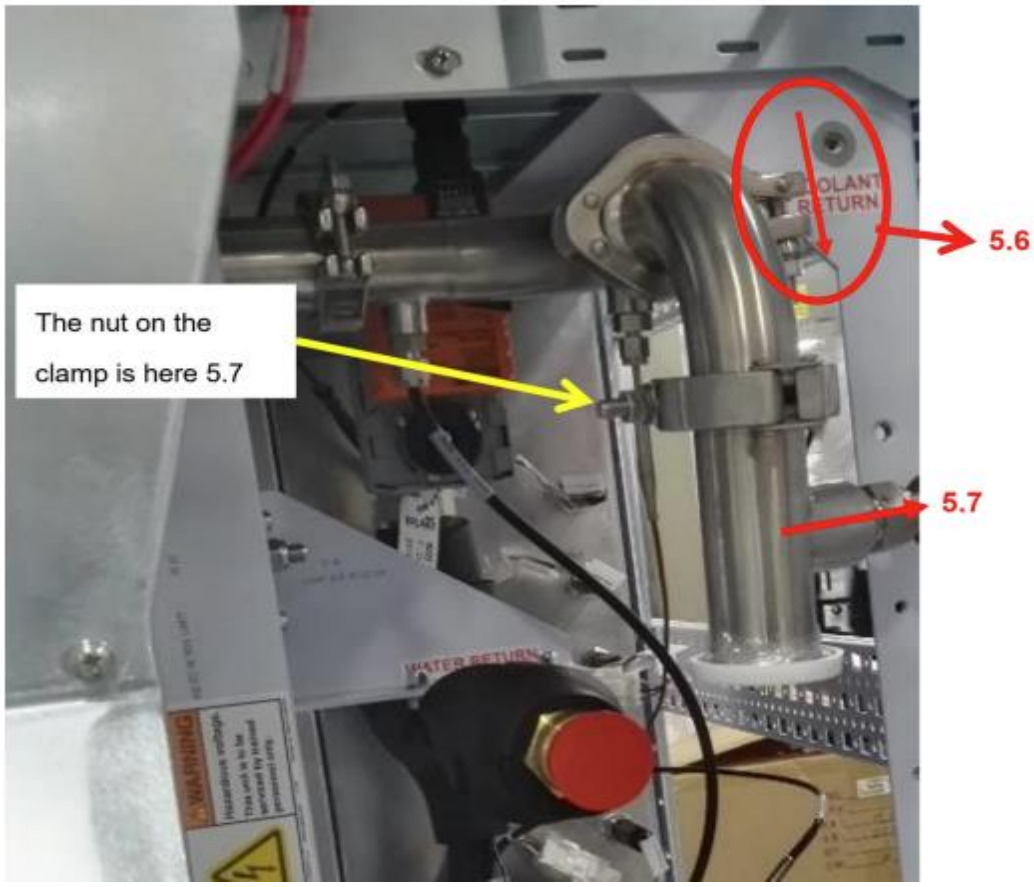
5.6 Install the 90° elbow (361-883-00) on the CDU cooling return pipe as shown in the figure. Use 1 clip (361-882-00) and washer (361-886-00) fixed.

Note: The outlet direction of the outlet 90° elbow is downward as shown in the figure. The direction of the nut on the clamp is toward the lower right position.

5.7 Install the T-tube (633-220-00) on the previous 90° elbow as shown in the figure. Use 1 clip (361-882-00) and washer (361-886-00) fixed.

Note: The exit direction of the T-tube is to the right as shown in the figure. The direction of the clamp nut is toward the left position on the back.

Note: Do not use torque to fix the clip. The T-shaped tube needs to be loosened when the tube is installed in the T-shaped fitting.



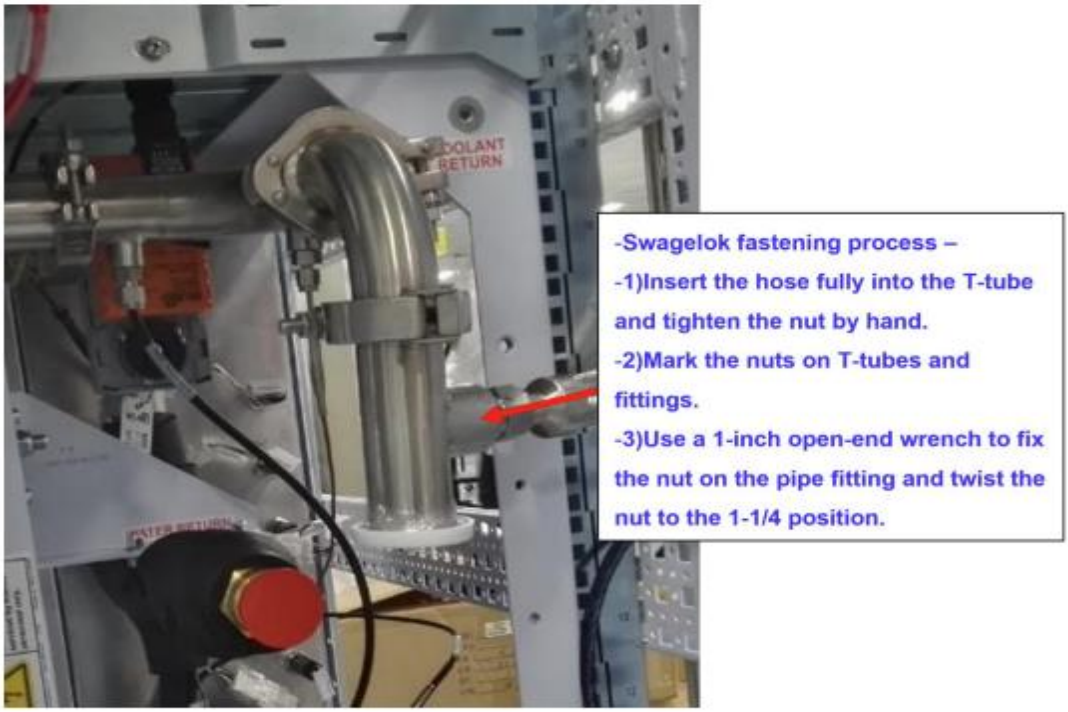
MAN: :CERTIFIED OPERATOR, TECHNICIAN

MATERIAL: N/A

MACHINE: N/A

METHOD:

5.8 Insert the hose into the T-tube as shown.
a. Rotate the 90° elbow and T-tube direction as required so that the hose can be inserted into the pipe fitting.
b. Confirm that the hose is fully inserted into the fitting.
5.9 Manually tighten the nuts on the pipe fittings.
5.10 Use a 13mm socket wrench to fix the nut on the clamp with a torque value of 3.3 N·m.
5.11 Mark the nuts on T-tubes and fittings.
5.12 Use a 1-inch open-end wrench to fix the nut on the pipe fitting, and twist the nut to the 1-1/4 position.



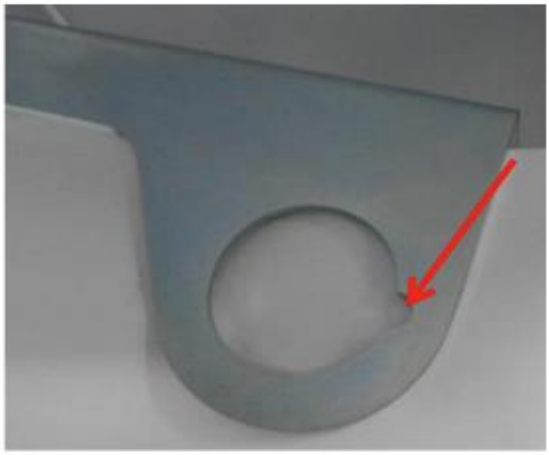
MAN: :CERTIFIED OPERATOR, TECHNICIAN

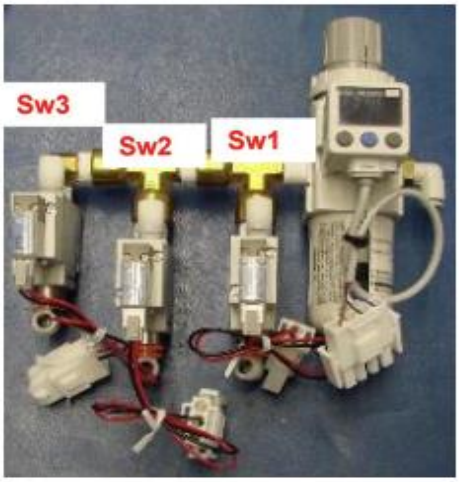
MATERIAL: N/A

MACHINE: N/A

METHOD:

6.2 Remove the plastic nut on the regulator (653-632-00) as shown in the figure.
6.3 Install the voltage regulator on the bracket as shown in the figure-when installing the positioning PIN on the voltage regulator into the notch of the bracket-install from below and fix it with a plastic nut.





MAN: :CERTIFIED OPERATOR, TECHNICIAN

MATERIAL: N/A

MACHINE: N/A

METHOD:

6.1 As shown in the figure, install a bracket (634-569-00) to the frame at the lower right corner of the cabinet maintenance surface. Use 2 M8 frame locking blocks (614-766-90)
And the following materials, the tool is a 5mm hexagon socket head, the torque value is 13.5 N·m.

a. 2 M8 screws (470-933-01).
b. 2 M8 spring pads (470-502-07).
c. 2 M8 flat washers (470-501-06).

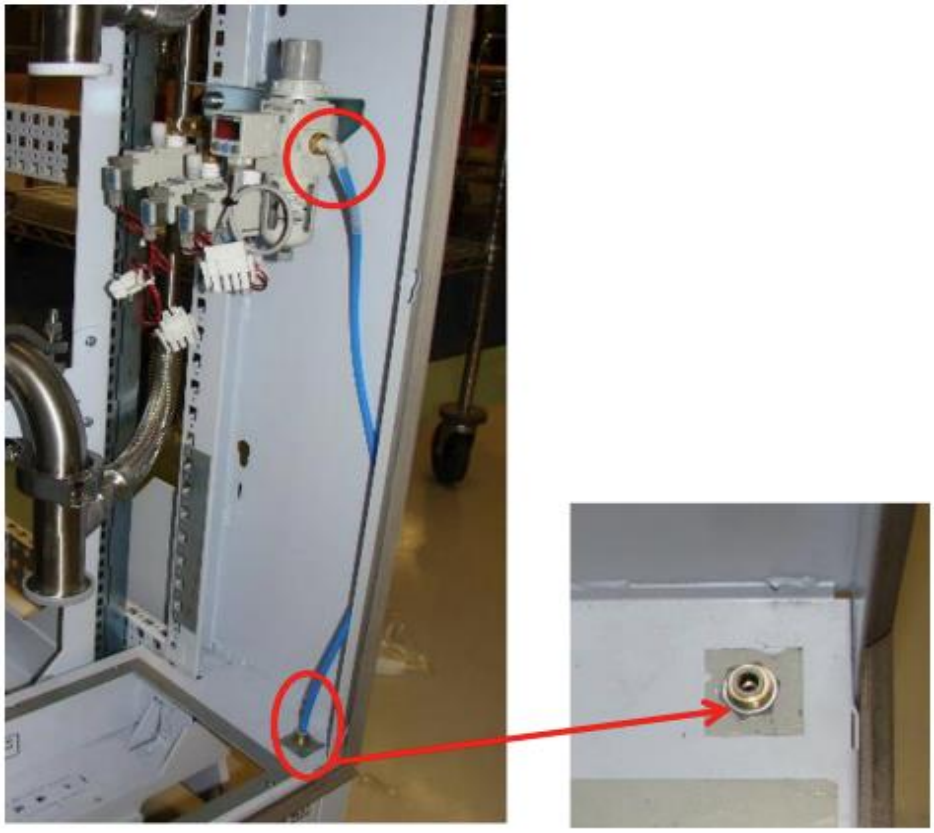
MAN: :CERTIFIED OPERATOR, TECHNICIAN

MATERIAL: N/A

MACHINE: N/A

METHOD:

6.4 Install 1 Dixon connector (637-717-00) under the air conditioner at the bottom of the cabinet as shown in the figure. The tool used is a 17mm socket wrench, The torque value is 5.6 N·m.
6.5 As shown in the figure, install a 560mm air pipe (637-746-01) with one end to the input end of the regulator, and the other end to the Dixon connector.



MAN: :CERTIFIED OPERATOR, TECHNICIAN

MATERIAL: **N/A**

MACHINE: **N/A**

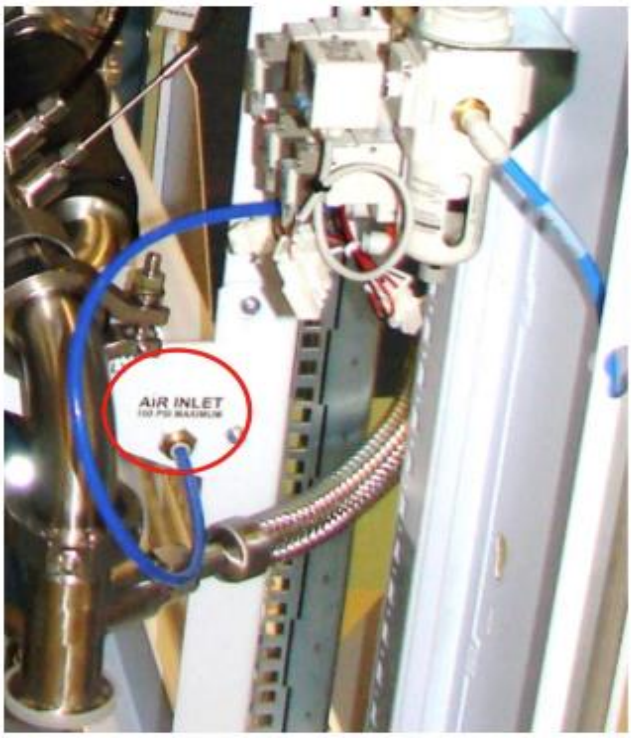
METHOD:

6.6 Take a CDU interlocking cable (634-441-00) as shown in the figure, connect one end to the ADB-P72 interface, and the other end to the air conditioner SW2 interface.
Place the extra wire on the frame rail and fix it with a cable tie (813-929-01).

MAN: :CERTIFIED OPERATOR, TECHNICIAN

MATERIAL: **N/A**

MACHINE: **N/A**

METHOD:

6.7 Insert one end of a 500mm air pipe (637-746-00) into the outlet of the SW2 air conditioner and the other end into the CDU port as shown in the figure.